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Hydraulic tool having ram piston with integrated overload assembly

Abstract

Embodiments of the invention provide a ram assembly for a hydraulic tool. The ram assembly includes a ram piston having a ram cavity. The ram piston receives a hydraulic pressure reaction force. The ram assembly further includes an overload assembly disposed in the ram cavity or in a cavity defined by a manifold. The overload assembly includes a burst disc positioned at a first end of the ram cavity or manifold, a lock nut positioned at a second end of the ram cavity or manifold, and a spacer positioned between the burst disc and the lock nut. The spacer transfers a supporting force applied to the lock nut to the burst disc. The hydraulic pressure reaction force and the supporting force both act in the same direction, and thus, are additive forces acting on the burst disk.

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Background/Summary

RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/051,396, filed Oct. 28, 2020, now U.S. Pat. No. 11,999,042, which represents the national stage entry of International Application No. PCT/US2020/048357, filed Aug. 28, 2020, which claims priority to U.S. Provisional Application No. 62/893,607, filed Aug. 29, 2019, entitled “Hydraulic Tool Having Ram Piston Design with Integrated Overload Assembly,” the content of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Hydraulic crimpers and cutters are different types of hydraulic power tools, such as portable, handheld hydraulic tools, for performing work (e.g., crimping or cutting) on a work piece. A hydraulic pump pressurizes hydraulic fluid and transfers it to a cylinder in the tool. This cylinder causes an extendible piston to be displaced toward a cutting or crimping head. The piston exerts a force on the head of the power tool, which typically includes opposed jaws with certain cutting or crimping features, depending upon the particular configuration of the power tool. In this case, the force exerted by the piston closes the jaws to perform cutting or crimping on a work piece (e.g., a wire) at a targeted location.

(2) One known hydraulic tool can include an overload assembly configured to burst if the hydraulic tool exceeds a predetermined high-pressure set point. In normal operation, when the hydraulic tool reaches or exceeds the predetermined high-pressure set point, a load-sensing device of the hydraulic power tool can shut down a motor of the hydraulic tool. If the load-sensing device fails to shut off the motor at the predetermined high-pressure set point, the overload assembly can burst, opening high pressure lines to a reservoir and preventing the hydraulic tool from pressurizing. A typical overload assembly can include a lock nut that is in contact with a spacer, which separates the lock nut from a burst disc (also referred to as a “burst cap”) or similar overload device.

(3) There are certain perceived disadvantages of using an assembly such as this, however. For example, during operation of the hydraulic tool, downward movement of the piston pressurizes the hydraulic fluid and forces the hydraulic fluid into the hydraulic fluid passage circuit, causing a reaction force to push on the burst disc, which in turn causes a supporting force from the lock nut to counter the reaction force from the hydraulic pressure. However, because the two forces are in opposite directions, the resulting force that is required to seal the burst disc decreases, which can result in leakage at the burst disc. In order to achieve a significantly larger resulting force, the supporting force on the burst disc must increase, reducing the fatigue life of the burst disc and working against the sealing of the burst disc.

SUMMARY

(4) Embodiments of the invention provide a ram assembly for a hydraulic tool. The ram assembly

includes a ram piston having a ram cavity at an end of the ram piston, the ram piston configured to receive a hydraulic pressure reaction force and an overload assembly. The overload assembly can be disposed in the ram cavity. The overload assembly can include a burst disc positioned at a first end of the ram cavity and a lock nut positioned at a second end of the ram cavity, the second end opposite the first end, and configured to receive a supporting force. The hydraulic pressure reaction force and the supporting force can be additive.

(5) In some embodiments, the hydraulic tool can include a manifold that defines a ram chamber having a fluid inlet configured to receive a pressurized hydraulic fluid, a ram assembly having a ram piston disposed within at least a portion of the ram chamber, a burst disc having a peripheral flange, a spacer in contact with the burst disc at the peripheral flange, and a lock nut configured to receive a first force acting in a first direction and a second force acting in the first direction.

(6) Some embodiments of the invention provide a hydraulic tool that can include a manifold defining a ram chamber and a cavity. The cavity can be located along a hydraulic fluid path within the hydraulic tool. The hydraulic tool can include an overload assembly disposed in the cavity defined by the manifold. The overload assembly can include a burst disc positioned at a first end of the cavity, and a lock nut positioned at a second end of the cavity, the second end opposite the first end. During operation of the hydraulic tool, hydraulic pressure can be created in the ram chamber, by which a hydraulic pressure reaction force acts on the lock nut and a supporting force from the lock nut acts on the burst disc so that both the hydraulic pressure reaction force and the supporting force act in the same direction and seal the burst disc against the first end of the cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and form a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of embodiments of the invention:

(2) FIG. 1 is a cross-sectional view of an overload assembly that exists in some known hydraulic tools.

(3) FIG. 2 is a cross-sectional view of an overload assembly according to one embodiment of the invention.

(4) FIG. 3 is an exploded cross-sectional view of a portion of the overload assembly of FIG. 2.

(5) FIG. 4 is a cross-sectional view of a portion of the overload assembly according to another embodiment of the invention.

(6) FIG. 5 is an exploded cross-sectional view of a portion of the overload assembly of FIG. 4.

(7) FIG. 6 is an exploded perspective view of an overload assembly according to another embodiment of the invention.

(8) FIG. 7 is a cross-sectional view of the overload assembly of FIG. 6.

DETAILED DESCRIPTION

(9) The following discussion is presented to enable a person skilled in the art to make and use embodiments of the invention. Various modifications to the illustrated embodiments will be readily apparent to those skilled in the art, and the generic principles herein can be applied to other embodiments and applications without departing from embodiments of the invention. Thus, embodiments of the invention are not intended to be limited to embodiments shown, but are to be accorded the widest scope consistent with the principles and features disclosed herein. The following detailed description is to be read with reference to the figures, in which like elements in different figures have like reference numerals. The figures, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of embodiments of the invention. Skilled artisans will recognize the examples provided herein have many useful

alternatives and fall within the scope of embodiments of the invention.

(10) As used herein, unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings.

(11) The overload assembly according to embodiments of the invention can be part of a hydraulic power tool. In one embodiment, the hydraulic power tool can include a cutting or crimping head, an electric motor, a pump driven by the motor, and a housing defining a cylinder. An extendable ram piston can be disposed within the cylinder. The pump can provide pressurized hydraulic fluid through a hydraulic fluid passage circuit to the ram piston, causing the ram piston to extend from the housing to actuate the jaws of the cutting or crimping head for cutting or crimping a work piece, such as a wire. Other power sources can be used to power the tool. Once a work piece or other target is placed between the jaws, the hydraulic power tool can be powered to close the jaws to perform a cutting or crimping action and cut or crimp the work piece or other target.

(12) As discussed above, known hydraulic power tools can include an overload assembly that bursts when the hydraulic tool exceeds a predetermined high-pressure set point, such as when a primary pressure control device (e.g., a pressure transducer) of the hydraulic tool fails to shut off the motor at the predetermined high-pressure set point.

(13) FIG. 1 is an example of an overload assembly **100**. The overload assembly **100** and its components are housed in a manifold **102**. Components of the overload assembly **100** include a lock nut **104**, a spacer **106**, and a burst disc **108**. The overload assembly **100** is positioned proximate to a portion of a hydraulic fluid passage circuit **110**. In the overload assembly **100**, the lock nut **104** counters a hydraulic pressure reaction force **112** that pushes on the burst disc **108** with a supporting force **114**.

(14) An increase in the hydraulic pressure reaction force **112** acting on the burst disc **108** can reduce the sealing force on the burst disc **108** against the mounting surface **116**. Additionally, because the supporting force **114** counteracts the hydraulic pressure reaction force **112**, an increase in the hydraulic pressure reaction force **112** induces an increase in the supporting force **114** acting on the burst disc **108**, causing fatigue of the burst disc **108**.

(15) Accordingly, the overload assembly according to embodiments of the invention integrates a burst disc and other overload assembly components into a ram piston of a hydraulic tool, creating forces during operation that are additive instead of opposing. In some embodiments, the overload assembly can be integrated into a manifold of a hydraulic tool so that additive forces are created.

(16) FIG. 2 illustrates an overload assembly **200** according to one embodiment of the invention. The overload assembly **200** can be housed within a manifold **202** defining a ram chamber **201** configured to contain various component parts of a ram assembly, such as a ram piston **204** and a spring **206**. More particularly, the overload assembly **200** can be integrated with the ram assembly, namely, with the ram piston **204**. For example, one end of the ram piston **204** can include a ram cavity **208** in which a lock nut **210**, a spacer **212**, and a burst disc **214** can be disposed. In particular, the end of the ram piston **204** that includes the ram cavity **208** can be the end of the ram piston **204** located proximate to a fluid inlet **216** through which the ram chamber **201** is in fluid communication with other portions of the hydraulic fluid passage circuit.

(17) In some embodiments, the spacer **212** includes an aperture **213**. In some embodiments, the spacer **212** includes a peripheral flange **215** extending generally radially. Although the spacer **212** is included in the overload assembly **200**, alternative embodiments of the overload assembly **200** might not include a spacer. Further, in other embodiments, additional components could be included between lock nut **210** and the burst disc **214** additionally or alternatively to the spacer **212**.

(18) The spring **206** can surround an outer surface of the ram piston **204**. In some embodiments, the spring **206** can be positioned to extend from a front portion **203** of the ram chamber **201** to a back

portion **205** of the ram chamber **201** during cutting or crimping actions. The spring **206** can be affixed at the front portion **203** of the ram chamber **201**. In some embodiments, the ram chamber **201** might contain another type of device instead of a spring **206**, such as an O-ring, for example.

(19) The lock nut **210** can be configured and arranged so that a supporting force created by the lock nut **210** (i.e., supporting force **218**, which is a force generated by the torqueing of the lock nut **210**) acts in the same direction as a hydraulic pressure reaction force (i.e., a hydraulic pressure reaction force **220**) that pushes on the ram piston **204** (and thus pushes on the lock nut **210**).

(20) As shown in FIG. 2, the burst disc **214** is located at the first end **226** of the ram cavity **208**, the lock nut **210** is positioned at a second end **228** of the ram cavity **208**, opposite the first end **226**. In addition, the lock nut **210** can be in threaded contact with an interior surface of the ram cavity **208** or can be coupled to the ram cavity **208** in an alternative manner. Further, the spacer **212** is positioned between the burst disc **214** and the lock nut **210**.

(21) FIG. 3 illustrates a portion of the overload assembly **200** of FIG. 2. In particular, FIG. 3 shows, from left to right, the ram cavity **208** of the ram piston **204**, the burst disc **214**, the spacer **212**, and the lock nut **210**. Although no threading or similar structure is shown on an interior surface of the ram cavity **208**, some embodiments of the overload assembly **200** can have the lock nut **210** in threaded contact with the interior surface of the ram cavity **208**.

(22) In operation of a hydraulic tool that includes the overload assembly **200**, hydraulic fluid passes through the fluid inlet **216** and creates hydraulic pressure at the back portion **205** of the ram chamber **201**, creating the hydraulic pressure reaction force **220** that facilitates movement of the ram piston **204**. Further, the supporting force **218** acts on the burst disc **214** (i.e., by being transmitted by the spacer **212**) in the same direction as the hydraulic pressure reaction force **220**, as shown in FIG. 2. Since both the supporting force **218** and the hydraulic pressure reaction force **220** are acting in the same direction, the two forces are additive and both act on the burst disc **214**. Thus, both forces work to seal the burst disc **214** against a mounting surface of the ram cavity **208** (i.e., mounting surface **221**).

(23) Having the overload assembly **200** integrated into the ram piston **204** in this manner can advantageously utilize forces applied during operation of the hydraulic tool to help seal the burst disc **214**, even at higher pressures, without causing excessive force to be placed on the burst disc **214**. This can advantageously help achieve an improved sealing of the burst disc **214**. Since less force is placed on the burst disc **214**, the fatigue life of the burst disc **214** can be lengthened.

(24) In alternative embodiments, a similar sealing-assistance effect can be achieved with the overload assembly **200** in alternative locations. Particularly, instead of being integrated in the ram piston **204**, the overload assembly **200** can be positioned in the manifold **202** in such a way (e.g., having a particular orientation and location in the manifold **202**) that causes the hydraulic pressure reaction force **220** and the supporting force **218** to be additive.

(25) FIG. 4 illustrates an alternative embodiment where the overload assembly **200** is located in the manifold **202** proximate to the back portion **205** of the ram chamber **201**. More particularly, the overload assembly **200** is located along a portion of a hydraulic fluid path that defines a fluid outlet **222** through which the ram chamber **201** is in fluid communication with another location, such as a fluid reservoir of the hydraulic tool. To facilitate this, a portion of the manifold **202** along the hydraulic fluid path (and, in this particular arrangement, along the portion of the hydraulic fluid path that defines the fluid outlet **222**) can be configured (e.g., machined) to include a bore **230** or other type of cavity that houses the overload assembly **200**. As shown, the burst disc **214** can be sealed against a mounting surface **223** of the manifold **202** at one end of the cavity (i.e., opposite the other end of the cavity that is located closer to the back portion of the ram chamber **201**).

(26) In operation of a hydraulic tool that includes the overload assembly **200** shown in FIG. 4, hydraulic fluid passes through the fluid inlet **216** and creates hydraulic pressure at the back portion **205** of the ram chamber **201**, creating the hydraulic pressure reaction force **220** in the ram chamber **201**, which acts on the lock nut **210**. Further, the supporting force **218** acts on the burst disc **214**

(i.e., by being transmitted by the spacer **212**) in the same direction as the hydraulic pressure reaction force **220**. Since both the supporting force **218** and the hydraulic pressure reaction force **220** are acting in the same direction, the two forces are additive and both act on the burst disc **214**. Thus, both forces work to seal the burst disc **214** against mounting surface **223** of the ram cavity **208**.

(27) There are other perceived disadvantages of using known hydraulic tools as well, such as known hydraulic tools that include the overload assembly **100** of FIG. **1** or similar overload assemblies. For example, in some known overload assemblies, such as the overload assembly **100** of FIG. **1**, the lock nut **104** and the spacer **106** are typically selected so that the surfaces of the lock nut **104** and the spacer **106** that are in contact with each other are flat or substantially flat. In this arrangement, however, the lock nut **104** and the spacer **106** can become misaligned, such as when the contacting surfaces are not machined to a desired degree (e.g., a gap exists between the lock nut **104** and the spacer **106** at one or more locations across their contacting surface areas) and/or when one or both components are displaced due to movement during normal operation of the hydraulic tool. This misalignment can, in turn, create or increase a radial imbalance in the sealing force on the burst disc **108**, in which case the burst disc **108** might not be held down properly enough against the manifold **102** to sufficiently keep the burst disc **108** sealed in place.

(28) In some embodiments of the invention, the lock nut **210** and the spacer **212** of the overload assembly **200** can be configured to help self-align during operation of the hydraulic tool. To facilitate this, each of the lock nut **210** and the spacer **212** can have radially-contoured surfaces **224**, **225** that allow the overload assembly **200** to compensate for misalignment that might result during operation of the hydraulic tool or for other reasons. For example, as shown in FIG. **2**, and as similarly shown in FIGS. **3**, **4**, and **5**, a surface **224** of the lock nut **210** can be a convex radial surface and a surface **225** of the spacer **212** can be a concave radial surface. In some embodiments, surface **224** and surface **225** can be contoured so that a radius of surface **224** substantially matches a radius of surface **225**, which can help promote alignment between the two surfaces so that they have substantially matching contours. Alternative configurations of surface **224** and/or surface **225** are possible as well. For example, surface **225** can be a conical surface instead of a concave radial surface. As another example, surface **224** can be a concave radial surface and surface **225** can be a convex radial surface.

(29) In these embodiments, a substantially constant force can be maintained against the burst disc **214**, preventing or reducing force on the burst disc **214** and keeping the burst disc **214** (i.e., the peripheral flange **215** of the burst disc **214**) in place flat against the surface to which it is mounted (i.e., mounting surface **221**).

(30) FIG. **6** illustrates an exploded view of an overload assembly **300** according to another embodiment of the invention, and FIG. **7** illustrates a cross-sectional view of the overload assembly **300**. In particular, the overload assembly **300** includes a lock cap **302**, a ball **304**, a spacer **306**, and a burst disc **308**.

(31) The lock cap **302** can be configured to lock the overload assembly **300** in a manifold (e.g., the ram cavity **208**) and to provide a force on the ball **304**. The lock cap **302** can include a recess **305** at one end of the lock cap **302**, where the recess **305** is configured to house at least a portion of the ball **304**. The recess **305** can be contoured to substantially match a contour of the ball **304**.

(32) The lock cap **302** can be made of metal or another material. In alternative embodiments, the lock cap **302** can take the form of a lock nut that is threaded to another surface (e.g., to the manifold **102** or to the ram cavity **208**), such as lock nut **210** of FIG. **2**.

(33) The ball **304** can be a spherical object made of metal or another material. The ball **304** can be configured to support radial and/or axial loads and transfer loads from the lock cap **302** to the spacer **306**. The ball **304** can act as a universal joint for alignment of the overload assembly **300**.

(34) In alternative embodiments, the lock cap **302** and the ball **304** can be integrated together by machining a sphere on a bottom end of the lock cap **302**. In these embodiments, the bottom end of

the lock cap **302** can include a spherical protrusion configured to contact the spacer **306** and transfer loads from the lock cap **302** to the spacer **306**, and the spacer **306** can include a recess **307** that is contoured to substantially match a contour of the spherical protrusion.

(35) The spacer **306** can be configured to house at least a portion of the ball **304** so that the spacer **306** receives the load from the ball **304**. For example, the spacer **306** can include a recess **307** that is contoured to substantially match a contour of the ball **304**. Further, the spacer **306** can be configured to serve the same or similar purpose as the spacer **212** of FIG. 2 (i.e., to transfer forces to the burst disc **308** and to provide a leak path when the burst disc **308** is ruptured). The burst disc **308** can take the same or similar form and serve the same or similar purpose as the burst disc **214** of FIG. 2.

(36) With the arrangement of the overload assembly **300**, the ball **304** helps maintain a substantially balanced force against the spacer **306** (and thus, against the burst disc **308**) during operation of the hydraulic tool, improving the effectiveness of the seal of the burst disc **308** (e.g., keeping the peripheral flange **215** of the burst disc **308** in place flat against mounting surface **221** of FIG. 2, or mounting surface **223** of FIG. 4). Furthermore, the overload assembly **300** (and, likewise, the embodiments having the concave/convex/etc. surfaces described above) can be cheaper and easier to manufacture.

(37) By the term “substantially” or “about” used herein, it is meant that the recited characteristic, parameter, value, or geometric planarity need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

(38) The description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. An overload assembly for use in a ram cavity of a ram piston of a hydraulic tool, the ram cavity including a first end and a second end opposite the first end, the overload assembly comprising: a burst disc positioned at the first end of the ram cavity; a lock nut positioned at the second end of the ram cavity, the lock nut receiving from the ram piston a hydraulic pressure reaction force acting in a first direction, the lock nut receiving a supporting force acting in the first direction; and a spacer positioned between the burst disc and the lock nut.
2. The overload assembly of claim 1, wherein the spacer transfers the supporting force applied from the lock nut to the burst disc.
3. The overload assembly of claim 1, and further comprising a ball positioned between the lock nut and the spacer.
4. The overload assembly of claim 1, wherein the burst disc includes a radially extending peripheral flange that contacts a mounting surface of the first end.
5. The overload assembly of claim 1, wherein the lock nut is threaded to the second end of the ram cavity and the supporting force is generated by torqueing the lock nut.
6. The overload assembly of claim 1, wherein the spacer includes a concave radial surface and the lock nut includes a convex radial surface, the concave radial surface being in contact with the convex radial surface and having matching contours.
7. The overload assembly of claim 1, and further comprising a lock cap to lock the overload assembly in a manifold.

8. The overload assembly of claim 1, wherein the spacer supplies a leak path when the burst disc is ruptured.
9. A hydraulic tool comprising: a manifold including a ram chamber that receives a pressurized hydraulic fluid; a ram assembly having a ram piston, the ram piston receiving a hydraulic pressure reaction force, the ram piston including a ram cavity within at least a portion of the ram chamber, the ram cavity including a mounting surface; a burst disc sealed against the mounting surface; a lock nut that receives the hydraulic pressure reaction force and a supporting force acting in a same direction as the hydraulic pressure reaction force; and a spacer positioned between the burst disc and the lock nut to transfer the supporting force from the lock nut to the burst disc.
10. The hydraulic tool of claim 9, wherein the hydraulic pressure reaction force results from the pressurized hydraulic fluid pushing on the ram piston during operation of the hydraulic tool.
11. The hydraulic tool of claim 9, and further comprising a hydraulic fluid passage circuit to hydraulically drive the hydraulic tool; and a fluid inlet in fluid communication with the hydraulic fluid passage circuit.
12. The hydraulic tool of claim 9, wherein a spring surrounds an outer surface of the ram piston.
13. The hydraulic tool of claim 9, wherein the lock nut and the spacer have radially-contoured surfaces to self-align during operation.
14. The hydraulic tool of claim 9, wherein the lock nut is threaded to the ram cavity.
15. A ram assembly for a hydraulic tool, the ram assembly comprising: a ram piston that receives a hydraulic pressure reaction force, the ram piston having a ram cavity; and an overload assembly in the ram cavity, the overload assembly including: a burst disc; a lock nut that receives a supporting force, the hydraulic pressure reaction force and the supporting force acting in a same direction; and a spacer positioned between the burst disc and the lock nut.
16. The ram assembly of claim 15, wherein the spacer transfers the supporting force from the lock nut to the burst disc.
17. The ram assembly of claim 15, wherein the burst disc includes a peripheral flange that seals the burst disc against the ram cavity.
18. The ram assembly of claim 15, wherein the spacer includes a concave radial surface and the lock nut includes a convex radial surface.
19. The ram assembly of claim 15, wherein the overload assembly is housed within the ram piston.
20. The ram assembly of claim 15, and further comprising a spring that surrounds an outer surface of the ram piston.
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